

KAAN BERK CAN

Game & Systems Designer

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PROFILE

Game designer with an industrial engineering background, most at home with systems, economies and balance. I spent three years on Turkish Airlines' crew-planning software, mainly optimization and reporting, and I now bring the same modeling and simulation work to games. Current projects include Dekrawler, a card roguelike on itch.io that I balance with Monte Carlo runs, and two team prototypes I led in Godot and Unity. I'm finishing an MSc in Game and Interaction Technologies at ITU.

SKILLS

Design	Systems and economy design · progression and balancing · technical design · level and puzzle design · GDDs and specs
Analysis	Mathematical modeling · Monte Carlo simulation · game analytics · Python · SQL · R · Excel
Engines & tools	Unity (C#) · Godot · Unreal Engine 5 · Git · Jira · Tabletop Simulator
Languages	Turkish (native) · English (C1) · German (A1) · Japanese (A1)

GAME PROJECTS

Dekrawler ↗

Unity 6 · Solo · v0.4.0 on itch.io

A roguelike where the dungeon is a deck of cards and your weapon degrades to the rank of whatever it last killed. Thirteen campaign levels, 45 jokers bound to individual cards and 15 artifacts that change the rules. Balance is checked with Monte Carlo runs against the real game build. The interface is drawn in code and all sound is synthesized at runtime, so the build ships without art or audio files.

Idle Incrementation Analysis ↗

Unity · Excel · Python · Solo

A 15-minute idle economy with six upgrade tiers, each purchase costing 10% more than the last, modeled second by second in Excel. Human players beat both greedy and brute-force Python optimizers because they were willing to hold gold for a later payoff. Playable on PC and Android.

Harvey Park ↗

Godot · 4-person team, lead · Alpha demo

First-person horror about noticing what changed. You memorize a room in the dark, check it again once the power is back, and decide in the elevator whether something is wrong. Sixteen floors, a hidden timer and a pursuer that only moves when you look away.

Birth of Miracle ↗

Unity · 2–3 person team, lead · Vertical slice

2.5D puzzle platformer told mostly through its environment. Timing and logic puzzles, abilities that open up earlier areas, and a notebook that lets players decide how far into the lore they want to go.

Miracle: Born & Dead ↗

Design document · Solo · 6+ months

Strategy game design with empire turns on a campaign map and real-time battles between five asymmetric factions.

Tabletop designs

Board & megagames

Miracle: Board & Game, a co-op RPG for 1–6 players on a 13×13 grid. Moods of Istanbul, a negotiation game for 4 to 16+ players built on Istanbul IPA statistics and playable in Tabletop Simulator. Mikapoly, a two-player co-op reworking of Monopoly with no elimination.

Game concepts

Pitch documents

Sown in Silence, a farming sim that turns into cosmic horror at night; Gears of Behemoth, about climbing and taking apart mechanical giants; Dissonance Ward, procedural hospital horror built around shifting personalities.

Player & design research

Papers · video essays

First-author paper comparing human and generative-AI creativity in game design. A survey study on how age, gender and play frequency shape what players think. Video essays on Settlers of Catan and on Path of Exile through Bartle's player types.

EXPERIENCE

Independent Game Developer & MSc Researcher

2024 – Present

Istanbul, Turkey

- Thesis question: can the emotional weight of a text survive once it is reduced to game mechanics? I extract procedural mechanics from a text and simulate them with everything else stripped away.
- Led the four-person team behind Harvey Park and a smaller one on Birth of Miracle. Solo work in Unity and Godot, and regular game jams.

Turkish Airlines · Crew Planning / Systems Development

Aug 2022 – Jul 2025

Istanbul, Turkey

Administrator / System Developer

Oct 2024 – Jul 2025

- Split my time between development on the Jeppesen crew suite (Pairing, Rostering, Tracking, Manpower) and system and machine management in Bash, Oracle and MS SQL.
- Part of the team that moved the codebase from Mercurial to Git.

System Developer

Apr 2023 – Oct 2024

- Rewrote the assignment format and constraints of the crew assignment optimizer. Runs got 10–25% faster, and planners had fewer rule violations to fix by hand.
- Wrote Rave rules, Python tools and SQL, Power BI and Excel VBA reports used to plan more than 20,000 cockpit and cabin crew.
- Took requests straight from crew planners and turned them into rules and reports, tracked in Jira.

Intern Developer (part-time)

Aug 2022 – Apr 2023

- Started part-time on the same team while finishing the internships my degree required.

EDUCATION

MSc, Game and Interaction Technologies · Istanbul Technical University

2024 – exp. Dec 2026

BSc, Industrial Engineering (English) · Yıldız Technical University

2019 – 2023

GPA 3.46 / 4.00